



## HEAT PUMPS

All the currents near the equator flow westward, then turn north and south in western boundary currents such as the Gulf Stream and Brazil Current. These carry warm water to cooler regions, making their winters milder. Meanwhile, eastern boundary currents such as the Humboldt Current and the Benguela Current carry cool polar water into the tropics.



### ▲ TROPICAL GARDENS

*The Gulf Stream gives the Scilly Isles in the north Atlantic a surprisingly warm climate.*

## VISIBLE FLOW

Where warm and cold currents meet, the cold water pushes below the warm water, stirring up minerals from the sea floor that are vital for tiny drifting algae called plankton. This fuels plankton growth, providing food for fish and other animals. This effect is most marked in shallow continental shelf seas, because the seabed is nearer the surface. Sometimes the two currents are made visible by different-colored plankton blooms.

### ◀ COLOR-CODED

*Seen from space, these plankton blooms mark where the warm Brazil Current (carrying the blue plankton) meets the colder Falklands Current (carrying the green plankton).*